

Perturbation Pipeline

Design overview for the multilingual fact-checking robustness study

1. Why We Need a Perturbation Pipeline

Our paper investigates the robustness of embedding models used in fact-checking pipelines — specifically for claim matching and evidence retrieval — when the input text is perturbed. Real-world claims on social media are noisy: they contain typos, OCR artefacts, hashtags, dialect shifts, missing hedge words, and adversarial formatting. We need a systematic way to generate all of these variations at scale across ten languages so we can measure how much each perturbation type degrades retrieval performance.

The pipeline takes a claim text and a language code, applies every perturbation, and returns a dictionary of perturbed versions ready for embedding and evaluation.

2. Architecture

The pipeline has two layers. Layer 1 is our own perturbation set, which runs on all ten languages. Layer 2 is a replication of the perturbations from Pikuliak et al. (ACL 2025 Findings), which is applied to English inputs only — this enables a direct comparison with the paper's results.

Layer	Coverage	Count	Method
Our perturbations	EN, DE, ES, FR, PL, HBS, HI, AR, ML, ZH	14	LLM (7) + Rule-based (7)
Paper perturbations	EN only	12	LLM (10) + Rule-based (2)
Total (EN input)	—	26	—
Total (non-EN input)	—	14	—

3. Our Perturbation Set (All Languages)

Each perturbation targets a different real-world noise source observed on social media. LLM-based perturbations use a local Qwen3.6-35B-A3B model served via LM Studio with thinking disabled and structured JSON output, achieving ~1.3 s per call. Rule-based perturbations run in pure Python with no external calls.

ID	Family	Description	Method
A1_emoji_relevant	Social media	Append 2 topic-relevant emojis	LLM
A1_emoji_disruptive	Social media	Insert 1 jarring emoji mid-text	Rule
A2_hashtagification	Social media	3 keyword hashtags + 3 spam tags appended	LLM
A3_ocr_artifacts	Social media	3 OCR character confusions (o→0, e→3, a→@)	Rule
A4_stt_artifacts	Social media	3 phonetic homophone substitutions	LLM

ID	Family	Description	Method
B1_qualifier_removal	Semantic	Strip all hedge / qualifying language	LLM
B2_temporal_drift	Semantic	Shift dates +3 years + modern anchor phrase	LLM
C1_homoglyphs	Typography	3 Cyrillic/Greek lookalike substitutions	Rule
C2_leetspeak	Typography	3 leet-speak character replacements	Rule
C3_word_splitting	Typography	2 adversarial spaces inside keywords	LLM
D2_clickbait_llm	Style	Clickbait prefix + 3 ALL-CAPS words (LLM)	LLM
D2_clickbait_rule	Style	Clickbait prefix + caps (A/B: rule-based)	Rule
D3_back_trans_de	Style	EN → DE → EN round-trip drift	Rule
D3_back_trans_es	Style	EN → ES → EN round-trip drift	Rule
E2_presupposition	Pragmatic	Loaded presupposition frame prepended	LLM

4. Paper Perturbation Set (English Only)

These replicate the taxonomy from Pikuliak et al. (ACL Findings 2025), which benchmarked embedding models on multilingual previously fact-checked claim retrieval. Including them lets us place our results directly on the same scale as the paper's reported numbers.

ID	Category	Description	Method
P_typos_low	Typos	1 char-level typo (min Levenshtein)	Rule
P_typos_high	Typos	3 char-level typos (max Levenshtein)	Rule
P_negation_low	Negation	Single negation of main claim	LLM
P_negation_high	Negation	Double negation (not true that ... not ...)	LLM
P_entity_low	Entity swap	Replace 1 named entity with plausible alt.	LLM
P_entity_high	Entity swap	Replace all named entities	LLM
P_llm_rewrite_low	LLM rewrite	Minimal paraphrase (1–2 word changes)	LLM
P_llm_rewrite_high	LLM rewrite	Extensive rewrite, same meaning	LLM
P_dialect_aae	Dialect	Rewrite in African American English (AAVE)	LLM
P_dialect_jamaican	Dialect	Rewrite in Jamaican Patois	LLM
P_dialect_pidgin	Dialect	Rewrite in Nigerian Pidgin English	LLM
P_dialect_singlish	Dialect	Rewrite in Singlish	LLM

5. Language Coverage

Ten languages are tested, organized into three Indo-European family clusters and one typological stress sample. Each language's LLM-based perturbations use a language-specific prompt with native hedge words, phoneme examples, clickbait phrases, and presupposition frames. Languages without a specific prompt fall back to the English instruction.

Stratum	Languages	Rationale
C1 — Germanic	English (EN), German (DE)	Baseline floor. Same script, different morphology.
C2 — Romance	Spanish (ES), French (FR)	Latin script, grammatical gender, verb morphology.
C3 — Slavic	Polish (PL), Serbo-Croatian (RS)	Latin script, case marking; typos have large semantic impact.
Stress Sample	Hindi (HI), Arabic (AR), Malayalam (ML), Mandarin (ZH)	Four orthogonal typological probes: Abugida script, +Abjad, agglutination, no-space word boundaries.

6. Technical Setup

LLM calls go through the OpenAI-compatible API of LM Studio running locally. Three settings are configured at the server level (not in code):

Setting	Value	Reason
Model	Qwen3.6-35B-A3B Q4_K_M	Best local multilingual model available
Thinking mode	Disabled	97 % of tokens were reasoning; 44× slower with no quality gain
Response format	JSON Schema {perturbed_text: string}	Eliminates formatting noise, enforces clean output
Temperature	0.7	Some lexical variation without hallucination

Benchmark: with thinking disabled and structured output, average latency is 1.3 s per LLM call. All 26 English perturbations complete in under 15 seconds on a single claim.

7. Repository Files

File	Purpose
perturbations.py	Main pipeline — perturb_all(text, lang) returns all perturbations
perturbation_prompts.json	System prompt + per-perturbation prompts for all 10 languages
verify_perturbations.py	Verifier — checks validity and meaning preservation of outputs

Appendix — Full Prompt Catalogue

All prompts are stored in `perturbation_prompts.json`. English prompts are shown in full below. Language-specific prompts for DE, ES, FR, PL, HBS are shown inline. Prompts for HI, AR, ML, ZH contain Unicode text and are referenced by language code — open the JSON file to view them.

System Prompt (shared by all calls)

You are a text perturbation engine for a multilingual NLP research pipeline. Apply exactly the perturbation described in the user message. If the perturbation cannot be meaningfully applied, place the original text unchanged in the output field.

A — Our Perturbations (All Languages)

A1_emoji_relevant **SURFACE**

English

Append exactly 2 emojis to the end of this claim that are thematically relevant to its topic. Do not insert emojis inside the text body. Do not change any words.

A2_hashtagification **SURFACE**

English

Keep the original claim text exactly as written. Then append exactly 3 hashtag keywords derived from the most important nouns in the claim (e.g. #Government #Policy #ClimateChange) followed by exactly 3 generic spam hashtags (e.g. #BreakingNews #MustRead #Viral). Do not modify any word in the original sentence.

A4_stt_artifacts **SURFACE**

English

Replace exactly 3 words in this claim with phonetically similar real words that a speech-to-text system would confuse (e.g. 'their' → 'there', 'policy' → 'police', 'won' → 'one', 'weather' → 'whether', 'new' → 'knew', 'change' → 'chance', 'right' → 'write'). Every replacement must be a valid existing word — never a misspelling. Apply one replacement per word. You must always make at least 1 substitution — never return the original text unchanged.

German

Ersetze in diesem deutschen Text genau 3 Wörter durch klanglich ähnliche echte Wörter, die ein Spracherkennungssystem verwechseln würde (z.B. 'wieder' → 'wider', 'Meer' → 'mehr', 'war' → 'wahr', 'seit' → 'seid', 'das' → 'dass', 'stil' → 'still'). Jede Ersetzung muss ein gültiges deutsches Wort sein — niemals eine Falschschreibung. Ein Ersatz pro Wort. Falls weniger als 3 geeignete Wörter vorhanden sind, ersetze so viele wie

möglich.

Spanish

Reemplaza exactamente 3 palabras en este texto en español por palabras reales fonéticamente similares que un sistema de reconocimiento de voz confundiría (p.ej. 'echo' → 'hecho', 'tubo' → 'tuvo', 'bello' → 'vello', 'caza' → 'casa', 'coser' → 'cocer', 'hola' → 'ola'). Cada reemplazo debe ser una palabra española válida, nunca una falta de ortografía. Un reemplazo por palabra. Si hay menos de 3 candidatos válidos, aplica tantos como sea posible.

French

Remplace exactement 3 mots dans ce texte en français par des mots réels phonétiquement similaires qu'un système de reconnaissance vocale confondrait (ex. 'son' → 'sont', 'peu' → 'peut', 'ou' → 'où', 'ces' → 'ses', 'mer' → 'mère', 'ver' → 'vert'). Chaque remplacement doit être un mot français valide, jamais une faute d'orthographe. Un remplacement par mot. S'il y a moins de 3 candidats valides, applique autant que possible.

Polish

Zastąp dokładnie 3 słowa w tym polskim tekście brzmiącymi podobnie prawdziwymi słowami, które system rozpoznawania mowy mógłby pomylić (np. 'może' → 'morze', 'czas' → 'czasz', 'rzepa' → 'rzeka', 'bar' → 'bar'). Każde zastąpienie musi być poprawnym polskim słowem – nigdy błędem ortograficznym. Jedno zastąpienie na słowo. Jeśli jest mniej niż 3 odpowiednie słowa, zastąp tyle ile możliwe.

Serbo-Croatian

Zamijeni tačno 3 riječi u ovom tekstu stvarnim riječima koje zvuče slično i koje bi sistem za prepoznavanje govora mogao pobrkati (npr. 'sat' → 'sad', 'grad' → 'grād', 'luk' → 'lûk', 'pas' → 'paz'). Svaka zamjena mora biti valjana riječ – nikad pogrešno napisana. Jedna zamjena po riječi. Ako ima manje od 3 pogodnih riječi, zamijeni koliko god je moguće.

Hindi [Unicode — see JSON file]

Arabic [Unicode — see JSON file]

Malayalam [Unicode — see JSON file]

Mandarin [Unicode — see JSON file]

B1_qualifier_removal SEMANTIC

English

Remove all hedging and qualifying language from this claim to make it sound like a definitive statement of fact. Target words and phrases such as: allegedly, reportedly, according to, sources say, may, might, could, would, appears to, seems to, is said to, claims that, suggests. If no qualifying language is present, return the text unchanged.

German

Entferne alle abschwächenden und relativierenden Formulierungen aus dieser Behauptung, um sie wie eine definitive Tatsache klingen zu lassen. Zielwörter und -phrasen: angeblich, anscheinend, soll, könnte, würde, laut, nach Angaben von, Berichten zufolge, mutmaßlich, behauptet, offenbar, scheinbar, möglicherweise. Falls keine solchen Formulierungen vorhanden sind, gib den Text unverändert zurück.

Spanish

Elimina todo el lenguaje de cobertura y calificación de esta afirmación para que suene como una declaración definitiva de hecho. Palabras y frases objetivo: supuestamente, según, podría, aparentemente, al parecer, se dice que, presuntamente, según informes, parece que, se afirma que, sugiere. Si no hay lenguaje calificador, devuelve el texto sin cambios.

French

Supprime tout le langage de couverture et de qualification de cette affirmation pour la faire sonner comme une déclaration factuelle définitive. Mots et expressions cibles : soi-disant, prétendument, selon, apparemment, il semblerait que, d'après, indiquerait, pourrait, aurait, il paraît que, on dit que, suggère. Si aucun langage qualificatif n'est présent, retourne le texte inchangé.

Polish

Usuń wszystkie wyrażenia osłabiające i kwalifikujące z tego stwierdzenia, aby brzmiało jak definitywne stwierdzenie faktu. Docelowe słowa i frazy: rzekomo, podobno, jakoby, wedługo, może, mógłby, twierdzi się, ponoć, wydaje się, jak donoszą, sugeruje. Jeżeli nie ma języka kwalifikującego, zwróć tekst bez zmian.

Serbo-Croatian

Ukloni sve ograde i kvalifikatore iz ove tvrdnje kako bi zvučala kao definitivni iskaz činjenice. Ciljne riječi i fraze: navodno, tobože, prema, navodi se, mogao bi, čini se, tvrdi se, kako se izvještava, izgleda da, sugerira. Ako nema kvalifikatornog jezika, vrati tekst nepromijenjen.

Hindi [Unicode — see JSON file]

Arabic [Unicode — see JSON file]

Malayalam [Unicode — see JSON file]

Mandarin [Unicode — see JSON file]

B2_temporal_drift **SEMANTIC**

English

Shift every year or date mentioned in this claim forward by exactly 3 years to create a plausible but incorrect timeline. Then append exactly one short phrase anchoring the claim to a modern context (e.g. 'This resurfaced recently and is being widely shared.' or 'Experts confirmed this again last month.'). If no dates are present, only append the modern anchor phrase.

C3_word_splitting **SURFACE**

English

Select exactly 2 important content words or named entities in this claim and insert a single space inside each word at a natural-looking position to split it adversarially (e.g. 'vaccination' → 'vaccin ation', 'government' → 'govern ment'). The splits should look like plausible typos or formatting errors. Do not change any other words.

Mandarin [Unicode — see JSON file]

Arabic [Unicode — see JSON file]

Hindi [Unicode — see JSON file]

Malayalam [Unicode — see JSON file]

D2_clickbait_11m SURFACE

English

Rewrite this claim in a clickbait style. Prepend exactly one sensationalist hook phrase (e.g. 'EXPOSED:', 'BREAKING:', 'You won't believe this:', 'SHOCKING TRUTH:'). Then identify exactly 3 key nouns or verbs in the body and write them in ALL CAPS – leave every other word in its original case. Do not add, remove, or change any factual content.

German

Schreibe diese Behauptung im Clickbait-Stil um. Stell genau eine sensationalistische Einleitungssphrase voran (z.B. 'ENTHÜLLT:', 'BREAKING:', 'DAS WERDEN SIE NICHT GLAUBEN:', 'SCHOCKIEREND:'). Identifiziere dann genau 3 Schlüsselsubstantive oder -verben im Text und schreibe sie in GROSSBUCHSTABEN – alle anderen Wörter bleiben in ihrer ursprünglichen Schreibweise. Füge keine Fakten hinzu, entferne keine oder ändere keine.

Spanish

Reescribe esta afirmación en estilo clickbait. Antepone exactamente una frase de gancho sensacionalista (p.ej. 'REVELADO:', 'URGENTE:', 'NO LO CREERÁS:', 'IMPACTANTE:'). Luego identifica exactamente 3 sustantivos o verbos clave en el cuerpo y escríbelos en MAYÚSCULAS – deja todas las demás palabras en su caso original. No añadas, elimines ni cambies ningún contenido factual.

French

Réécris cette affirmation dans un style clickbait. Préfixe exactement une phrase d'accroche sensationnaliste (ex. 'RÉVÉLÉ :', 'CHOQUANT :', 'VOUS NE LE CROIREZ PAS :', 'URGENT :'). Identifie ensuite exactement 3 noms ou verbes clés dans le corps et écris-les en MAJUSCULES – laisse tous les autres mots dans leur casse d'origine. N'ajoute, ne supprime ni ne modifie aucun contenu factuel.

Polish

Przepisz to twierdzenie w stylu clickbait. Poprzedź dokładnie jedną sensacyjną frazę wprowadzającą (np. 'UJAWNIONO:', 'SZOKUJĄCE:', 'NIE UWIERZYSZ:', 'PILNE:'). Następnie zidentyfikuj dokładnie 3 kluczowe rzeczowniki lub czasowniki w treści i napisz je WIELKIMI LITERAMI – pozostaw wszystkie inne słowa w oryginalnej formie. Nie dodawaj, nie usuwaj ani nie zmieniaj żadnych faktów.

Serbo-Croatian

Prepiši ovu tvrdnju u stilu clickbaita. Dodaj na početak točno jednu senzacionalističku frazu (npr. 'OTKRIVENO:', 'ŠOKANTNO:', 'NEČETE VJEROVATI:', 'HITNO:'). Zatim identificiraj točno 3 ključne imenice ili glagola i napiši ih VELIKIM SLOVIMA – ostavi sve ostale riječi u izvornom obliku. Nemoj dodavati, uklanjati niti mijenjati činjenici sadržaj.

Hindi [Unicode — see JSON file]

Arabic [Unicode — see JSON file]

Malayalam [Unicode — see JSON file]

Mandarin [Unicode — see JSON file]

E2_presupposition SURFACE

English

Prepend exactly one short loaded presupposition frame to this claim that smuggles in an assumption as if it were already established fact. Examples: 'In what should come as no surprise,', 'As has long been known,', 'Confirming what many already suspected,', 'Despite repeated cover-ups,'. Choose the frame that fits the topic most naturally. Do not change the original claim text itself.

German

Stell dieser Behauptung genau einen kurzen, vorbelasteten Präsuppositionsrahmen voran, der eine Annahme als bereits feststehende Tatsache einschmuggelt. Beispiele: 'Wie es schon lange bekannt ist,', 'Was viele bereits vermutet haben,', 'Trotz wiederholter Vertuschungsversuche,', 'Wenig überraschend,'. Wähle den Rahmen, der am natürlichsten zum Thema passt. Ändere den originalen Behauptungstext nicht.

Spanish

Antepone exactamente un marco de presuposición cargado a esta afirmación que introduce una suposición como si ya fuera un hecho establecido. Ejemplos: 'Como ya se sospechaba,', 'Lo que muchos ya sabían,', 'A pesar de los encubrimientos,', 'Sin que deba sorprender a nadie,'. Elige el marco que encaje más naturalmente con el tema. No cambies el texto original de la afirmación.

French

Fais précéder cette affirmation d'exactly un court cadre de présupposition chargé qui fait passer une hypothèse comme un fait déjà établi. Exemples : 'Comme on le sait depuis longtemps,', 'Confirmant ce que beaucoup soupçonnaient,', 'Malgré les tentatives de dissimulation,', 'Sans grande surprise,'. Choisissez le cadre qui correspond le plus naturellement au sujet. Ne modifie pas le texte original de l'affirmation.

Polish

Poprzedź to twierdzenie dokładnie jednym krótkim, obciążonym ramem presupozycji, która przemyca założenie jako już ustalony fakt. Przykłady: 'Jak od dawna wiadomo,', 'Potwierdzając to, co wielu podejrzewało,', 'Pomimo wielokrotnych prób zatuszowania,', 'Co nikogo nie powinno dziwić,'. Wybierz ramę, która najnaturalniej pasuje do tematu. Nie zmieniaj oryginalnego tekstu twierdzenia.

Serbo-Croatian

Dodaj na početku ove tvrdnje točno jedan kratak okvir napunjene presupozicije koji uvodi pretpostavku kao već utvrđenu činjenicu. Primjeri: 'Kao što je odavno poznato,', 'Potvrđujući ono što su mnogi sumnjali,', 'Unatoč ponovljenim prikrivanjima,', 'Što ne bi trebalo nikoga iznenaditi,'. Odaberi okvir koji se najprirodnije prilagođava temi. Ne mijenjaj originalni tekst tvrdnje.

Hindi [Unicode — see JSON file]

Arabic [Unicode — see JSON file]

Malayalam [Unicode — see JSON file]

Mandarin [Unicode — see JSON file]

B — Paper Perturbations (English Only)

Replication of the perturbation taxonomy from Pikuliak et al. (ACL Findings 2025). These are only applied when lang == 'EN'.

P_negation_low SEMANTIC EN only

English

Add a single negation to this claim so that it asserts the opposite of the original. Insert 'not', 'never', or an equivalent negation word at the most natural grammatical position. The result must be a fluent English sentence. Do not change any other words.

P_negation_high SEMANTIC EN only

English

Apply a double negation to this claim. First negate the core assertion, then wrap the whole thing in a denial frame. The output must follow this structure: 'It is not true that [single-negated version of the claim].' The result must be fluent English.

P_entity_low SEMANTIC EN only

English

Identify the single most prominent named entity in this claim (person, organisation, or location) and replace it with a different but plausible entity of the same type (e.g. replace one university with another university, one politician with another politician, one country with a neighbouring country). Keep all other words exactly as written.

P_entity_high SEMANTIC EN only

English

Identify every named entity in this claim (persons, organisations, locations, dates) and replace each one with a different but plausible entity of the same type. Keep all non-entity words exactly as written.

P_llm_rewrite_low SEMANTIC EN only

English

Rewrite this claim as a paraphrase that changes as few words as possible. You must change at least 1 word – do not return the original text. Swap one or two words for close synonyms or restructure one short phrase while keeping everything else identical. Preserve the exact meaning and all factual content.

P_llm_rewrite_high SEMANTIC EN only

English

Rewrite this claim using as many word changes as possible while fully preserving the meaning and all factual content. Aim for maximum Levenshtein distance from the original – use different vocabulary, restructure the

sentence, change word order. The rewritten claim must convey exactly the same information.

P_dialect_aae SURFACE EN only

English

Rewrite this claim in African American English (AAVE). Apply characteristic AAVE grammatical and lexical features such as habitual 'be', copula deletion, double negatives, and idiomatic vocabulary. Preserve all factual content exactly.

P_dialect_jamaican SURFACE EN only

English

Rewrite this claim in Jamaican Patois (Jamaican Creole). Apply characteristic Patois features such as 'mi' for 'I/me', 'dem' for 'them/those', 'nuh' for negation, 'fi' for 'to/for', and typical Jamaican lexical items. Preserve all factual content exactly.

P_dialect_pidgin SURFACE EN only

English

Rewrite this claim in Nigerian Pidgin English. Apply characteristic Naija Pidgin features such as 'e' for 'he/she/it', 'dem' for 'they', 'dey' for 'is/are/am', 'no' for negation, 'wey' as a relative pronoun, and typical Pidgin lexical patterns. Preserve all factual content exactly.

P_dialect_singlish SURFACE EN only

English

Rewrite this claim in Singlish (Singaporean English). Apply characteristic Singlish features such as discourse particles ('lah', 'leh', 'meh', 'lor', 'sia'), topic-fronting, copula omission, and typical Singlish expressions. Preserve all factual content exactly.